
Tutorial 1

Fundamentals of Computer Security

Computer Security

BCS2420

Part A

1. Which of the following is not one of the fundamental goals of computer security?
 - ☐ Confidentiality
 - ☐ Integrity
 - ☐ Repudiation
 - ☐ Availability
2. Which statement best characterizes the principle of authentication in computer security?
 - ☐ Ensuring data are unaltered except by authorized parties.
 - ☐ Ensuring only authorized users can access resources.
 - ☐ Confirming the identity or genuineness of an entity or data source.
 - ☐ Guaranteeing continuous operational readiness of systems and networks.
3. Which of the following adversary attributes deals primarily with the financial resources at an attacker's disposal?
 - ☐ Objectives
 - ☐ Methods
 - ☐ Capabilities
 - ☐ Funding Level
4. In threat modeling, which approach specifically uses a visual hierarchical structure where the root node is the overall attack goal and leaves represent methods to achieve it?
 - ☐ STRIDE

- ☐ Attack Trees
 - ☐ Diagram-Driven Modeling
 - ☐ Checklists
5. In the risk equation $R = T \times V \times C$, the term V refers to:
- ☐ The cost of a successful attack
 - ☐ The vulnerability level of a system
 - ☐ The probability of a threat occurring
 - ☐ The total number of adversaries involved
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Part B

1. **Real-World Incidents**: From the lecture examples (WannaCry, NotPetya, Stuxnet, TRITON), choose one and discuss:
 - The nature of the attack (e.g., malware type, exploit used)
 - Its broader impact (financial, operational, reputational)
 - Why it represents a significant turning point in cyber threats.
2. **Security Policy and Attacks** : Describe the relationship between a security policy and an attack. In your answer, explain how a policy violation can lead to a non-secure state and give one example of a practical policy from an organizational context.
3. **Threat Modeling Methods** : Compare two different approaches to threat modeling (e.g., Diagram-Driven vs. STRIDE). Clearly outline:
 - How each approach identifies threats
 - Their advantages and disadvantages for complex systems
4. **Insider vs. Outsider Threats** : Analyze the security implications of insiders compared to outsiders. Which scenario is potentially more dangerous, and how might security measures differ in addressing the two?
5. **Model-Reality Gaps** : The lecture discussed how invalid assumptions can undermine even the best security models. Give an example of how such a gap might arise in a cloud service context. Propose one strategy to mitigate this gap.

Part C

1. **Applying the Risk Equation:** A company uses an internal application that handles critical financial transactions.
 - **Threat probability, T :** 0.02 (2% chance of occurring each year)
 - **System vulnerability, V :** 0.7 (70% chance the attack will succeed if attempted)
 - **Potential cost of a successful attack, C :** € 5,000,000

(a) Calculate the **annual risk R** using the equation: $R = T \times V \times C$

(b) Briefly discuss *two* ways the organization might reduce R from **both** the **technical** and **procedural** standpoints.
2. **Comparing Security Controls:** A university estimates that a successful breach of its research data would cost € 2 million in damages. They can invest in **Option A** (improved firewalls) that reduces the system's vulnerability from 0.8 to 0.3, or **Option B** (advanced intrusion detection + staff training) that lowers the threat probability from 0.1 to 0.02.
 - Threat probability (original): $T = 0.1$
 - Vulnerability (original): $V = 0.8$
 - Potential cost: $C = €2,000,000$

(a). **Option A:** Calculate the new risk R_A .

(b). **Option B:** Calculate the new risk R_B .

(c). Which option yields a **lower** risk? Suppose both options cost the same. Which would you recommend and **why**?